agentAI: A Research-Data Platform for Physical and Embodied Agents

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A researcher states a goal; a planning model proposes the project and its agents, and the researcher corrects it. They then run in a sandbox, on data they harvest, take from a robot as it drives, or receive by upload. Unattended, a loop is only as good as what it reports about itself—ours reported success through rounds in which nothing changed. This sheet is that coordination, and the review that makes it checkable.

Weed Detection



The corpus exists to train the weed detector the laser robot needs. Two agents work it unprompted: one harvests and audits public imagery, the other trains. Over 150,000 labelled images.

Rover



A four-wheel skid-steer narrow enough for a crop row, carrying its own battery and compute. Every drive it opens becomes a governed dataset as it runs: camera, GPS and two inertial units, over one API key.

Laser Weed Control Robot



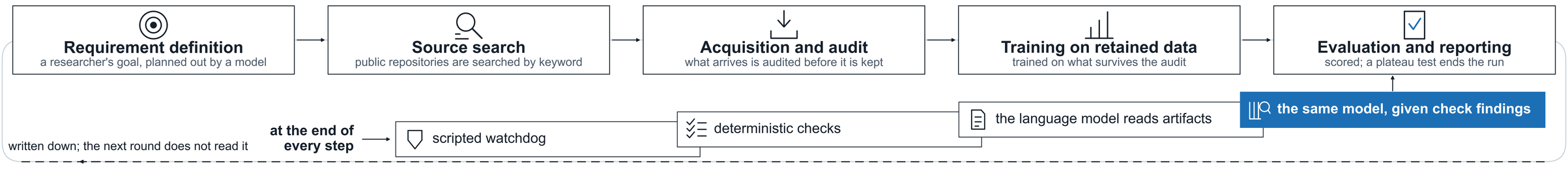
The end use the collection is for. A YOLO detector localises each weed and two galvanometer lasers aim at it and fire, destroying it in place—no chemical, and nothing touches the crop beside it.

Humanoid and Quadruped



Legs reach ground wheels cannot, and in the field this one crossed it faster than the cart. A wide stance and a level gait hold the camera steady, which is what a collector needs.

Four projects, one platform. Each was opened by the same procedure, and its data is registered the same way whether it was harvested, driven in by a robot, or uploaded.



Agent Workflow Diagram

Figure 1. The harvest-and-train loop, as it runs unattended. A researcher's goal is planned into keyword queries, what they return is audited, what survives is trained on, and the round is scored. The pictures that score the model also choose which epoch is reported, and the run ends on a plateau test rather than on a reviewer's judgement. Review follows every step, and the return edge is dashed because nothing reads the decision back yet.

ONE PLATFORM SUPPORTS ANY ROBOT OR DATASET

agentAI is one platform for every laboratory robot

A robot, drone or sensor rig is a project: its own dataset registry, its own agents, and an owner who controls both. Agents harvest, filter, label and train against that registry. Adding a robot is a configuration change.

Both the Rover and the laser robot transmit data

The Rover is a four-wheel skid-steer streaming camera, GPS and two inertial measurement units. The Laser Weed Control Robot carries two galvanometer lasers on independent channels and two cameras, one over the work zone and one forward.

Each drive becomes a dataset at its start

The platform registers a vehicle session as a governed dataset during the drive. Frames and sensor batches arrive over one API key. The laser robot transmits no position.

Every data type and domain forms its own project

On upload, a project returns an automatic diagnosis of the data before the next step is chosen. Weed detection is one project; the Rover and the laser robot are two others.

The platform is live; the QR code links to it

lab-b660m-c.tailfa6424.ts.net

Sign in with a Google account to open it.

OPEN-WEIGHT MODELS REVIEW IN TIERS

Citing evidence beats model size

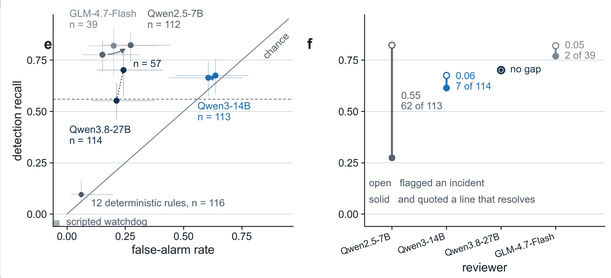


Figure 2. (e) Recall and (f) grounded recall on 149 frozen failures taken from this project's own record, scored by the project's own scorer. Each arrow runs from the model reading the raw files to the same model handed the twelve checks' findings. Grounded counts only the cases where the model also quoted a line that resolves the failure. The dotted arrow carries no head: its two ends were scored on different cases, 57 against 114.

We froze 149 real failures from our own history and asked open-weight models to find them. Good reviewers cite their evidence; size matters less. Handed the twelve checks' findings, a 27-billion model found 0.149 more, a 14-billion one 0.012.

Review runs even when nothing looks wrong

Review follows every step: failures here announced themselves as successes. Four levels by cost: the running watchdog, twelve fixed checks, a model reading files, then that model given the findings. The watchdog returned nothing on 149 frozen failures.

WHAT THE AGENTS DECIDED ON THEIR OWN

What the lab model says stays a draft

On the cluster	*the answer is authoritative*
plan the next step	qwen3.8 27B
review in depth	glm-4.7-flash
hard one-off reasoning	deepseek-v3 671B
decide what to collect	gemma 4
judge dataset quality	qwen2.5 7B
label assistance	minicpm-v

On the lab box	*the answer returns marked a draft*
turn intent into a project	qwen2.5 3B
summarise a dataset	qwen2.5 3B

filled square: the role exercises judgement

Figure 3. The eight jobs that need a model, and the model each one was given, read from the router's own table. A filled square marks a job that exercises judgement. The six on the cluster come back authoritative; the two in the lab come back marked a draft.

Eight jobs need a model. Most run on the cluster; two run in the lab on a 3B model, because a person is waiting. A lab answer is marked so it cannot count as a decision. Nine campaign reviews predate the mark: a 4.7 GB model wrote them while 458 GB sat idle.

The model's explanation failed its own test

Downloads carry no plant name, so every box from a site gets the same invented label. A model blamed that and wrote down first what would disprove it. A human approved; four runs disproved it. Web pictures average nine boxes, ours under two.

WHAT THE AGENTS DECIDED ON THEIR OWN (CONT.)

The search ran dry after round three

Two programs ran on the cluster unwatched, one searching for data, one training on it. They completed fifteen rounds. After the third the search found nothing new and the training set never changed, though the ledger and each round report logged new work.

WHAT THE COLLECTION PRODUCED

156,521 labelled by machine, 3,208 kept

A machine labelled 156,521 pictures; 59,134 remain once duplicate downloads are removed. We audited 13,527 of them, and only 3,208 came from a source worth keeping. The checker scores 1.000 on hand-labelled pictures, so a low score means the source is poor.

Gemma 4 said yes to every box

We cropped each box out and asked Gemma 4, a vision model, whether it held a weed. It said yes every time, wrong ones included, so it cannot serve as a check. Distance from trusted examples, on DINOv2 features, does the work. Fifty-one datasets were cut.

One bad box hides from a dataset filter

A labelling mistake is usually one bad box in an otherwise sound picture, invisible to a whole-dataset filter. The check must run box by box. We built those tools; their one run died on a bookkeeping bug and was never repeated, so we cannot say they work.